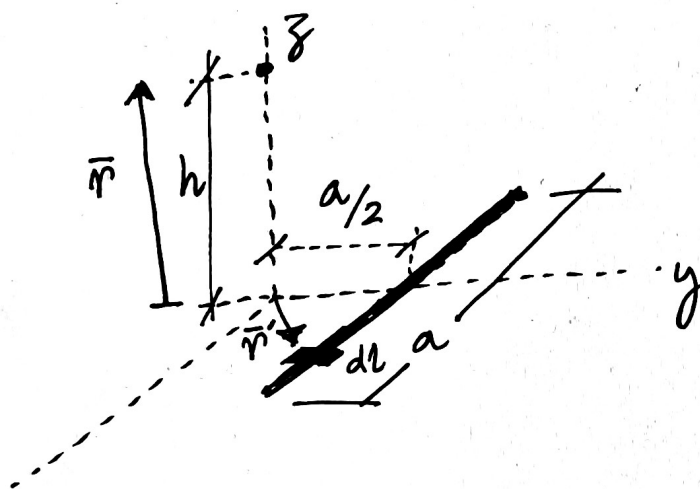


MIDTERM-1 SOLUTION.

Q1. Owing to symmetry, the electric field at point P contains only z-axis element. A possible solution is to consider any one of the sides, compute the field along z-axis and then multiply it by four. For the sake of simplicity we assume that the centre of the square-ring is at the origin.



$$\vec{r} = h \hat{a}_z$$

$$\vec{r}' = x' \hat{a}_x + y' \hat{a}_y$$

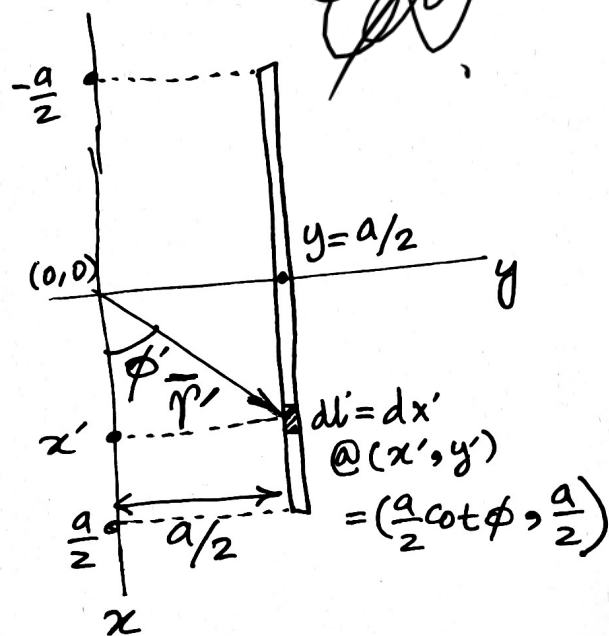
$$\vec{r}' = \frac{a}{2} \cot \phi' \hat{a}_x + \frac{a}{2} \hat{a}_y$$

$$dl' = dx' \quad (x' \text{ goes from } \frac{a}{2} \text{ to } -\frac{a}{2})$$

in other words, ϕ' goes from $+\frac{\pi}{4}$ to $\frac{3\pi}{4}$.

$$\vec{R} = \vec{r} - \vec{r}' = h \hat{a}_z - \frac{a}{2} \cot \phi' \hat{a}_x - \frac{a}{2} \hat{a}_y = h \hat{a}_z - x' \hat{a}_x - \frac{a}{2} \hat{a}_y.$$

$$R = |\vec{R}| = \left(h^2 + \frac{a^2}{4} \cot^2 \phi' + \frac{a^2}{4} \right)^{1/2} = \left(h^2 + x'^2 + \frac{a^2}{4} \right)^{1/2}$$



$$\vec{E}_1 = \int \frac{\rho_L dx' \vec{R}}{4\pi \epsilon_0 R^3} = \frac{\rho_L}{4\pi \epsilon_0} \int \frac{(h \hat{a}_z - \frac{a}{2} \hat{a}_y - x' \hat{a}_x) dx'}{\left(h^2 + x'^2 + \frac{a^2}{4} \right)^{3/2}}.$$

Since, we are interested in only z-component. (2)
we drop the terms involving Δx and Δy .

$$\bar{E}_1 = \frac{\rho_L h \Delta z}{4\pi\epsilon_0} \int_{x'=-\frac{a}{2}}^{+\frac{a}{2}} \frac{dx'}{(h^2 + \frac{a^2}{4} + x'^2)^{3/2}}$$

$$= \frac{\rho_L h \Delta z}{4\pi\epsilon_0} \left. \frac{x'}{(h^2 + \frac{a^2}{4}) \sqrt{h^2 + \frac{a^2}{4} + x'^2}} \right|_{-a/2}^{+a/2}$$

$$= \frac{\rho_L h \Delta z (\frac{a}{2} - (-\frac{a}{2}))}{4\pi\epsilon_0 (h^2 + \frac{a^2}{4}) \sqrt{h^2 + \frac{a^2}{4}}}$$

$$\bar{E}_1 = \frac{\rho_L h a \Delta z}{4\pi\epsilon_0 (h^2 + \frac{a^2}{4}) \sqrt{h^2 + \frac{a^2}{4}}}$$

$$\boxed{\bar{E}_{\text{net}} = 4\bar{E}_1 = \frac{\rho_L h a \Delta z}{\pi\epsilon_0 (h^2 + \frac{a^2}{4}) \sqrt{h^2 + \frac{a^2}{4}}}}$$

Q8 solution

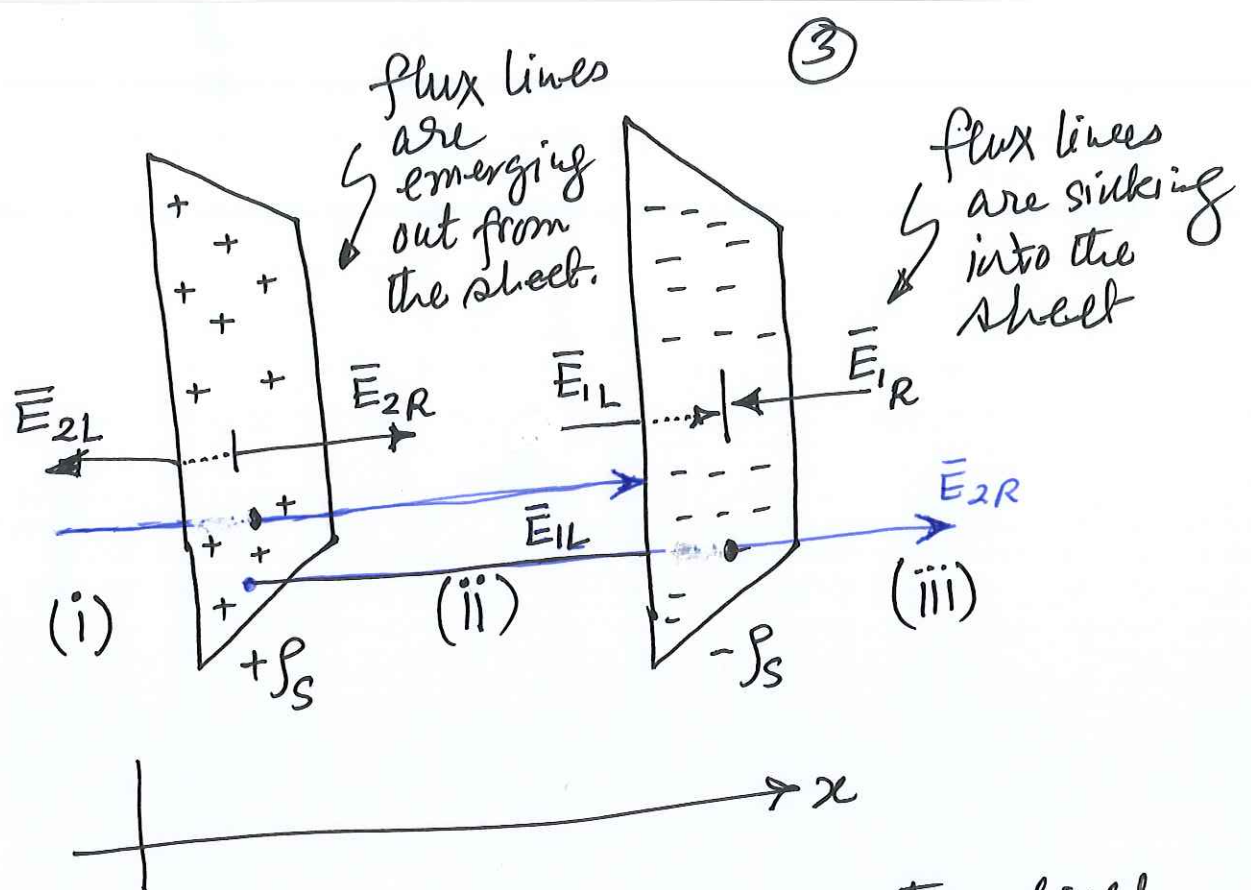
$$\text{Coulomb's Force of Repulsion} = \frac{q_A q_B}{4\pi\epsilon_0 a^2}$$

When string is cut, each point charge accelerate with Newton force equal to Coulomb's force of repulsion

$$\text{acceleration} = \frac{\text{Force}}{\text{mass}} = \frac{q_A q_B}{4\pi\epsilon_0 a^2 m}$$

put $m = m_A$ and $m = m_B$ to obtain the acceleration of charge q_A and q_B , respectively.

Q2.



For an infinite sheet of charge, the field is given as.

$$\vec{E} = \frac{\rho_s \hat{a}_N}{2\epsilon_0}$$

$\vec{E}_{2L}$ and $\vec{E}_{2R}$ are fields due to $+\rho_s$ on the left and right sides of the sheet.

$\vec{E}_{1L}$ and $\vec{E}_{1R}$ are fields due to $(-\rho_s)$ on the left and right sides of the sheet.

$$\vec{E}_{1R} = \frac{-\rho_s \hat{a}_x}{2\epsilon_0} \quad (\hat{a}_N = \hat{a}_x)$$

$$\vec{E}_{1L} = \frac{-\rho_s (-\hat{a}_x)}{2\epsilon_0} = \frac{\rho_s \hat{a}_x}{2\epsilon_0} \quad (\hat{a}_N = -\hat{a}_x)$$

$$\vec{E}_{2R} = \frac{+\rho_s \hat{a}_x}{2\epsilon_0} \quad (\hat{a}_N = \hat{a}_x)$$

$$\vec{E}_{2L} = \frac{+\rho_s (-\hat{a}_x)}{2\epsilon_0} = \frac{-\rho_s \hat{a}_x}{2\epsilon_0} \quad (\hat{a}_N = -\hat{a}_x)$$

$$\text{Field in region (i)} = \bar{E}_{2L} = \frac{-\rho_s \Delta x}{2\epsilon_0} + \frac{\rho_s \Delta x}{2\epsilon_0} = 0$$

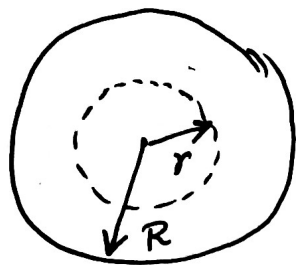
$$\text{Field in region (ii)} = \bar{E}_{2R} + \bar{E}_{1L} = \frac{\rho_s \Delta x}{\epsilon_0}$$

$$\text{Field in region (iii)} = \bar{E}_{1R} + \bar{E}_{2L} = \frac{-\rho_s \Delta x}{2\epsilon_0} + \frac{\rho_s \Delta x}{2\epsilon_0} = 0$$

Note that the field in ~~the~~ region (ii) is doubled due to facing both positive and negative charged surfaces.

This is exactly what happens in parallel plate capacitors. where field gets stronger in the region between electrodes and is zero outside the capacitors.

Q3.



(5)

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{enc.}}$$

at radius $r < R$, $Q_{\text{enc}} = \frac{4}{3}\pi r^3 \rho_v$

as the charge density is given to be uniform.

Due to symmetry, we assume $\vec{D} = D_r \hat{a}_r$.

field has only radial component.

$d\vec{S}$ is known to be $r^2 \sin\theta d\theta d\phi \hat{a}_r$.

$$\begin{aligned} \oint \vec{D} \cdot d\vec{S} &= \oint D_r r^2 \sin\theta d\theta d\phi = D_r r^2 \oint \sin\theta d\theta d\phi \\ &= 4\pi r^2 D_r. \end{aligned}$$

Comparing both sides, we get.

$$4\pi r^2 D_r = \frac{4}{3}\pi r^3 \rho_v$$

$$D_r = \frac{\frac{4}{3}\pi r^3 \rho_v}{4\pi r^2} = \frac{r \rho_v}{3}$$

$$\vec{D} = \frac{r \rho_v}{3} \hat{a}_r.$$

$$\boxed{\vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{r \rho_v \hat{a}_r}{3\epsilon_0}}$$

Q4.

(6)

(i) for $r < a$.

This region is empty. There is no free charge over here $Q_{enc} = 0$.

$$\oint \vec{D} \cdot d\vec{S} = \oint \epsilon_0 \vec{E} \cdot d\vec{S} = 0$$

$$\Rightarrow \vec{D} = 0 \text{ and } \boxed{\vec{E} = 0}$$

(ii) for $a \leq r \leq b$, we have

$$\rho_r = \frac{k}{r^3}$$

$$k \in \mathbb{R}$$

$$\int_V \equiv \int_{\tilde{r}=a}^r \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi}$$

Owing to symmetry, we assume.

$$\vec{D} = D_r \hat{a}_r$$

$$d\vec{S} = r^2 \sin\theta d\theta d\phi$$

$$\oint \vec{D} \cdot d\vec{S} = 4\pi r^2 D_r$$

$$Q_{enc} = \int_V \rho_r d\tau = \int_V \frac{k}{\tilde{r}^3} \tilde{r}^2 \sin\theta d\theta d\phi d\tilde{r}$$

$$= \int_V \frac{k}{\tilde{r}} d\tilde{r} \sin\theta d\theta d\phi = 4\pi k (\ln r - \ln a)$$

$$\Rightarrow D_r = \frac{4\pi k \ln(\tilde{r})|_a^r}{4\pi r^2} = k \frac{\ln(r) - \ln(a)}{r^2} = \frac{k \ln(\frac{r}{a})}{r^2}$$

$$\Rightarrow \boxed{E_r = \frac{D_r}{\epsilon_0} = \frac{k}{\epsilon_0} \frac{\ln(r/a)}{r^2}}$$

for $a \leq r \leq b$

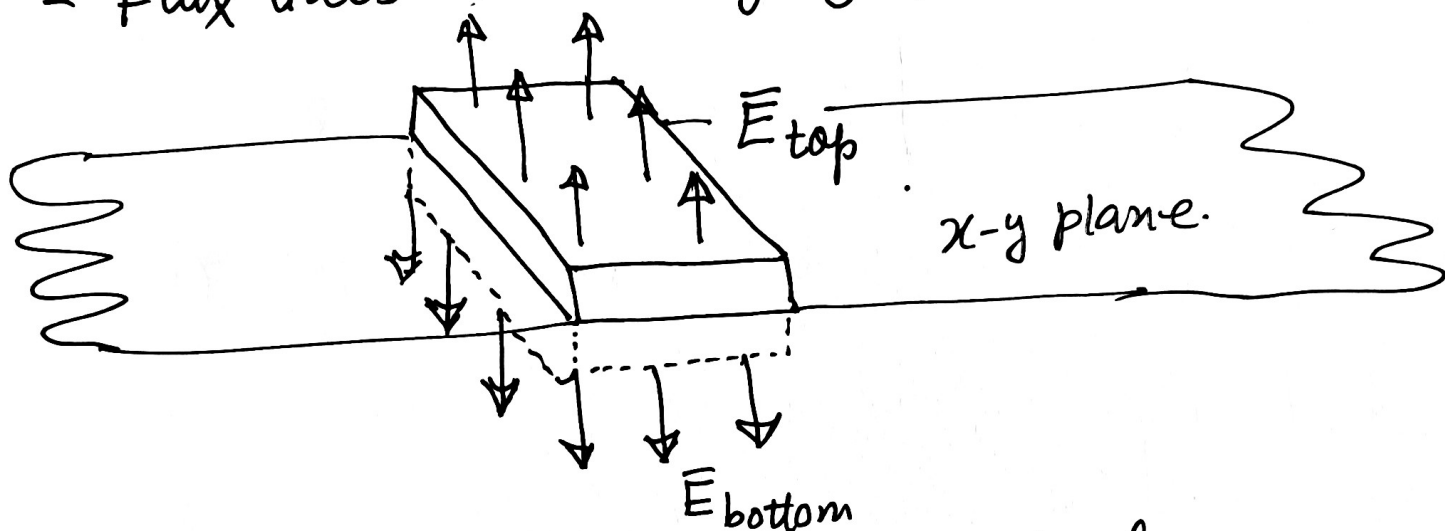
$$(iii) \text{ for } r > b, \quad \boxed{E_r = \frac{k}{\epsilon_0} \frac{\ln(b/a)}{r^2} \text{ for } r > b.}$$

Q5. - This is a tricky problem.

(7)

The Gaussian method may help solving this in a simpler manner.

- Let us assume the slab is positively charged.
- Flux lines are emerging from both sides.



- Due to symmetry, we realize that

$$\vec{E}_{top} = E_{top} \hat{z}$$

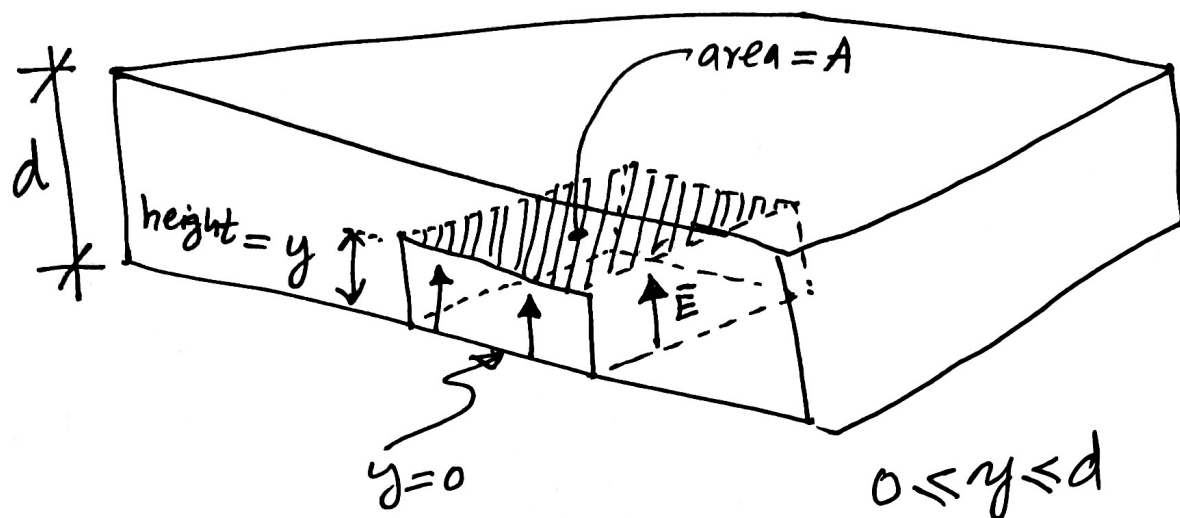
$$\vec{E}_{bottom} = E_{bottom} \hat{z}$$

$$\text{and } E_{bottom} = -E_{top}$$

(Equal in magnitude and opposite in direction.)

- Between the two directions of field, the value of the field at x-y plane must be zero.
- Consider a closed surface of height y and cross-sectional area A , the bottom of the surface is touching x-y plane.

(8)



$$\oint \epsilon_0 \vec{E} \cdot d\vec{S} = Q_{\text{enc.}}$$

$$\left. \begin{array}{l} \vec{E} = E_{\text{top}} \hat{z} \\ d\vec{S} = dS \hat{z} \end{array} \right\} \oint \epsilon_0 \vec{E} \cdot d\vec{S} = \epsilon_0 E_{\text{top}} \oint dS = \epsilon_0 E_{\text{top}} A$$

$$Q_{\text{enc}} = y A \rho_v$$

Equating both sides, we get

$$\epsilon_0 E_{\text{top}} A = y A \rho_v$$

$$E_{\text{top}} = \frac{y \rho_v}{\epsilon_0}$$

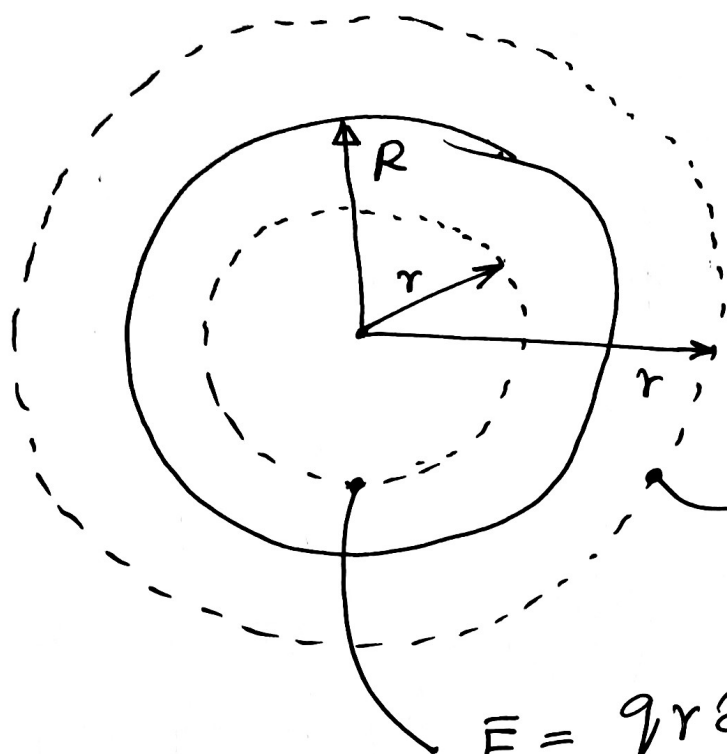
$$\vec{E}_{\text{top}} = \frac{y \rho_v}{\epsilon_0} \hat{z} \quad 0 \leq y \leq d$$

Similarly

$$\vec{E}_{\text{bottom}} = \frac{y \rho_v}{\epsilon_0} \hat{z} \quad -d \leq y \leq 0$$

Q6.

(9)



$$V(\infty) = 0 \text{ (given).}$$

$$\vec{E} = \frac{q dr}{4 \pi \epsilon_0 r^2} \text{ (outside)}$$

$$\vec{E} = \frac{q r dr}{4 \pi \epsilon_0 R^3} \text{ (inside)}$$

Potential outside the sphere: $d\vec{l} = dr \hat{r}$

$$V = - \int_{\infty}^{r(\text{outside})} \vec{E} \cdot d\vec{l} = - \int_{\infty}^r \frac{q dr}{4 \pi \epsilon_0 \tilde{r}^2}$$

$$= - \frac{q}{4 \pi \epsilon_0} \left(\frac{\tilde{r}^{-2+1}}{-2+1} \right) \Big|_{\infty}^r$$

$$= \frac{q}{4 \pi \epsilon_0} \frac{1}{\tilde{r}} \Big|_{\infty}^r = \frac{q}{4 \pi \epsilon_0 r} \quad (r > R)$$

Potential inside the sphere:

$$V = - \int_{\infty}^{r(\text{inside})} \vec{E} \cdot d\vec{l} = - \int_{\infty}^R \vec{E}_{\text{outside}} \cdot d\vec{l} - \int_R^r \vec{E}_{\text{inside}} \cdot d\vec{l}$$

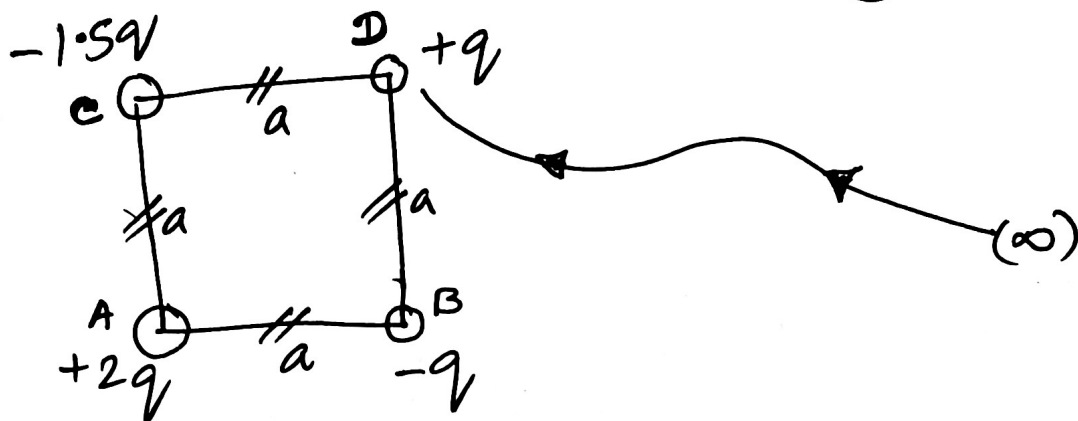
$$= \frac{q}{4 \pi \epsilon_0 R} - \frac{q}{4 \pi \epsilon_0 R^3} \frac{\tilde{r}^2}{2} \Big|_R^r = \frac{q}{4 \pi \epsilon_0 R} \left(1 - \frac{(r^2 - R^2)}{2R^2} \right)$$

$$= \frac{q}{4 \pi \epsilon_0} \cdot \frac{1}{2R} \left(3 - \frac{r^2}{R^2} \right) \text{ for } r < R$$

(10)

Q7.

(a)



The energy to bring $+q$ at position D is given as.

$$W_D = q_D V_D$$

$$= q_D (V_{DC} + V_{DA} + V_{DB})$$

$$= +q \left(\frac{(-1.5q)}{4\pi\epsilon_0 a} + \frac{(+2q)}{4\pi\epsilon_0 \sqrt{2}a} + \frac{(-q)}{4\pi\epsilon_0 a} \right)$$

$$= \frac{q^2}{4\pi\epsilon_0 a} \left(-1.5 + \frac{2}{\sqrt{2}} - 1 \right)$$

$$W_D = \frac{q^2 (\sqrt{2} - 2.5)}{4\pi\epsilon_0 a}$$

(11)

(b)

$$W_{\text{total}} = W_A + W_B + W_C + W_D$$

We bring charges in order A, B, C, and D.
 W_D has been evaluated already in (a).

$W_A = 0$ when field is empty.

$$W_B = (-q)V_B = (-q) \frac{(2q)}{4\pi\epsilon_0 a} = \frac{-2q^2}{4\pi\epsilon_0 a}$$

$$W_C = (-1.5q)V_C = (-1.5q) \left(\frac{2q}{4\pi\epsilon_0 a} + \frac{-q}{4\pi\epsilon_0 \sqrt{2}a} \right)$$

$$W_C = \frac{-1.5q^2}{4\pi\epsilon_0 a} \left(2 - \frac{1}{\sqrt{2}} \right)$$

$$W_D = \frac{q^2}{4\pi\epsilon_0 a} (\sqrt{2} - 2.5) \quad (\text{from part (a)})$$

$$W_{\text{total}} = \frac{q^2}{4\pi\epsilon_0 a} \left(-2 + \left(2 - \frac{1}{\sqrt{2}} \right) \left(-\frac{3}{2} \right) + (\sqrt{2} - 2.5) \right)$$

$W_{\text{total}} = \frac{-5.0251q^2}{4\pi\epsilon_0 a}$
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